

# Damiano Moscardini

M.Sc. Physics Student at University of Pisa

Website

<https://damianomoscardini.page>

GitHub

<https://github.com/damianomoscardini>



Email

[ciao@damianomoscardini.page](mailto:ciao@damianomoscardini.page)

LinkedIn

<https://linkedin.com/in/damianomoscardini>

## EDUCATION

---

### M.Sc. Physics (Quantum Computing and Technologies)

2025 - 2028 (Expected)

University of Pisa

### B.Sc. Physics

2021 - 2025

University of Pisa

### High School Diploma (Scientific)

2015 - 2020

Liceo Scientifico Antonio Vallisneri, Lucca

## SELECTED PROJECTS

---

### 3D-Printed Mach-Zehnder Interferometer

2026

Experimental Optics • Optomechanical Design • PyVisa • Arduino

A desktop-sized, 3D-printed Mach-Zehnder interferometer demonstrating wave interference. Fringe intensity is acquired via OPT101 photodiodes and a Siglent SDS824XHD oscilloscope, with the phase shift induced thermally via a Kapton heater.

→ <https://github.com/damianomoscardini/3d-printed-mzi>

### QSVM Encoding Benchmark

2026

Quantum Machine Learning • Python • Scikit-Learn • PennyLane

A benchmarking framework designed to compare quantum data-encoding strategies for QSVM kernels against a classical RBF-kernel SVM baseline, across datasets of varying geometry, using several dedicated metrics alongside standard accuracy.

→ <https://github.com/damianomoscardini/qsvm-encoding-benchmark>

# 3D-Printed "Single-Photon" Diffraction Experiment

2025

Experimental Quantum Optics • Photographic Detection • B.Sc. Thesis

A 3D-printed experimental setup demonstrating diffraction in the "single-photon" regime, using an attenuated laser and photographic film as a low-cost "single-photon" detector to record the pattern.

→ <https://github.com/damianomoscardini/3d-printed-single-photon-double-slit-diffraction>

## SKILLS

---

**Programming & Software:** Python • MATLAB • LabVIEW • LaTeX • Typst • AutoCAD

**Hardware & Tools:** 3D Printing • Arduino • Raspberry Pi

**Languages:** Italian (native) • English (C1 - First Certificate)

## EXPERIENCE

---

### Mathematics and Physics Tutor

2020 - Present

Provide continuous academic support for high school and university students. Develop personalized teaching methodologies and preparation strategies for national exams and university-level courses. Collaborated with Centro Yep! (Lucca, 2020 - 2023) and continued with private tutoring.

## CONFERENCES

---

### IEEE QCE26

Toronto, 2026

International Conference on Quantum Computing & Engineering | Attendee and Student Volunteer

### CISF26

Rome, 2026

Italian Conference of Physics Students | Attendee and Oral Presenter

### SIPE25

Bari, 2025

Spanish Italian Physics Exchange | Attendee and Oral Presenter

### LoT25

Pisa and Florence, 2025

Lights of Tuscany | Attendee and Poster Presenter

*Last updated: September 27, 2026*